

XuanLiang / XuanZhen Series Random Number Generator

Comprehensive User Manual

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Applicable Products	XuanLiang C01/W01, XuanZhen C01/W01
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1. Product Overview

The **XuanLiang series** (Quantum Random Number Generator) and the **XuanZhen series** (True Random Number Generator) are high-security random-number output devices designed around a physical entropy source. The products combine multiple physical noise sources with hardware cryptographic algorithms to output cryptographically secure random numbers compliant with the **GM/T 0005-2021** standard.

The device communicates with the host via USB and supports two protocols: **USB CDC** (virtual serial port) and **USB WinUSB** (native Bulk). It is compatible with mainstream operating systems including Windows, macOS and Linux, and can be widely applied in scenarios with strict random-number quality requirements such as cryptography research, secure communication, simulation, blockchain and financial encryption.

2. Model Reference

Market Name	Model	Communication Protocol	Description
XuanLiang C01	QS-QRNG-C01	USB CDC (Virtual Serial Port)	Quantum RNG, driver-free
XuanLiang W01	QS-QRNG-W01	USB WinUSB (libusb)	Quantum RNG, requires libusb
XuanZhen C01	QS-TRNG-C01	USB CDC (Virtual Serial Port)	True RNG, driver-free
XuanZhen W01	QS-TRNG-W01	USB WinUSB (libusb)	True RNG, requires libusb

Naming rule: the suffix **C01** indicates the CDC virtual-serial-port protocol; **W01** indicates the WinUSB protocol.

3. Host Software Download

Download the corresponding host program according to the device communication protocol:

Communication Protocol	Program Name	Download Method
USB CDC	rng_CDC.py	Scan QR code or visit the cloud drive
USB WinUSB	rng_WinUSB.py	Scan QR code or visit the cloud drive

Note: WinUSB-protocol users must also download libusb-1.0.dll (for Windows) and place it in the same directory as rng_WinUSB.py; otherwise the program will not run.

4. Software and Dependencies Overview

Protocol	OS	Host Program	Required Dependencies
USB CDC	Windows	rng_CDC.py	pyserial
USB CDC	macOS	rng_CDC.py	pyserial
USB CDC	Linux	rng_CDC.py	pyserial
USB WinUSB	Windows	rng_WinUSB.py	pyusb, libusb + libusb-1.0.dll
USB WinUSB	macOS	rng_WinUSB.py	pyusb, libusb (brew install libusb)
USB WinUSB	Linux	rng_WinUSB.py	pyusb, libusb (apt install libusb-1.0-0-dev)

5. Usage Instructions by Operating System

5.1 USB CDC Protocol Devices (XuanLiang C01 / XuanZhen C01)

Windows

1. Connect the device: Plug in the USB cable. Open Device Manager → Ports (COM and LPT) and check for the newly added USB Serial Device (COMx). If a yellow exclamation mark appears, right-click → "Update Driver" → "Search automatically".

2. Install dependencies:

```
pip install pyserial
```

3. Run the program:

```
python rng_CDC.py
```

Read data is automatically saved to the samples folder under the program directory.

macOS

1. Connect the device: Plug in the USB cable. The LED flashes 3 times and the device is ready.

2. Identify the port:

```
ls /dev/cu.usbmodem*
```

3. Install dependencies:

```
pip3 install pyserial
```

4. Run the program:

```
python3 rng_CDC.py
```

Data is automatically saved to the samples folder in the same directory.

Linux

1. Connect the device: Plug in the USB cable. The system usually auto-recognises it as /dev/ttyACM0.

2. Check the device:

```
ls /dev/ttyACM*
```

3. Permission setup (if access is denied):

```
sudo usermod -aG dialout $USER  
# Re-login for changes to take effect
```

4. Install dependencies:

```
pip3 install pyserial
```

5. Run the program:

```
python3 rng_CDC.py
```

Data is automatically saved to the samples folder in the same directory.

5.2 USB WinUSB Protocol Devices (XuanLiang W01 / XuanZhen W01)

Windows

1. Install WinUSB driver: Use the Zadig tool to replace the device driver with WinUSB.

2. Place the DLL: Copy `libusb-1.0.dll` (64-bit or 32-bit, matching your system) into the same directory as `rng_WinUSB.py`.

3. Install dependencies:

```
pip install pyusb libusb
```

4. Run the program:

```
python rng_WinUSB.py
```

Read data is automatically saved to the `samples` folder under the program directory.

macOS

1. Connect the device: Plug in the device.

2. Install libusb:

```
brew install libusb  
pip3 install pyusb libusb
```

3. Run the program:

```
python3 rng_WinUSB.py
```

Data is automatically saved to the `samples` folder in the same directory.

Linux

1. Connect the device: Plug in the device.

2. Install libusb:

```
sudo apt-get install libusb-1.0-0-dev  
pip3 install pyusb libusb
```

3. Permission setup (to avoid sudo every time):

```
sudo nano /etc/udev/rules.d/99-qs-qrng.rules  
# Add the following (replace actual VID/PID):  
SUBSYSTEM=="usb", ATTR{idVendor}=="xxxx", ATTR{idProduct}=="xxxx", MODE="0666"  
sudo udevadm control --reload-rules  
sudo udevadm trigger
```

4. Run the program:

```
python3 rng_WinUSB.py
```

Data is automatically saved to the `samples` folder in the same directory.

6. Command-Line Reading Examples

General Parameter Reference

Parameter	Description	Example
-s, --size	Read amount: 1M, 10M, 100M, 1G, 10G or byte count	-s 100M
-o, --output	Specify output file name	-o rand.bin
-p, --port	Manually specify port (CDC only)	-p COM3

By default, the program automatically saves data to the samples folder in the same directory. If -o is used to specify a file name, data is saved to the specified path.

CDC Protocol Examples (Windows / macOS / Linux)

```
:: Windows: read 100 MB
python rng_CDC.py -s 100M
:: Windows: manually specify COM port, read 1 GB
python rng_CDC.py -p COM3 -s 1G

# macOS: read 100 MB
python3 rng_CDC.py -s 100M
# macOS: manually specify serial port
python3 rng_CDC.py -p /dev/cu.usbmodemN32G45x1 -s 10M

# Linux: read 100 MB
python3 rng_CDC.py -s 100M
# Linux: specify port
python3 rng_CDC.py -p /dev/ttyACM0 -s 10M
```

WinUSB Protocol Examples (Windows / macOS / Linux)

```
:: Windows: read 10 MB
python rng_WinUSB.py -s 10M
:: Windows: read 1 GB with custom file name
python rng_WinUSB.py -s 1G -o myrand.bin

# macOS: read 1 GB
python3 rng_WinUSB.py -s 1G
# Linux: read 1 GB
python3 rng_WinUSB.py -s 1G
```

7. Performance Specifications

Communication Protocol	Output Rate	Description
USB CDC	Approx. 4.4 ~ 5.0 Mbps	USB Full-Speed virtual serial port measured stable value
USB WinUSB	Refer to specific model datasheet	Depends on physical entropy source and USB controller performance

CDC rate note: The theoretical upper limit of USB Full-Speed is approximately 12 Mbps. Affected by cryptographic operations, health checks and protocol overhead, CDC devices actually achieve a stable output of about **550 ~ 625 KB/s**, which is within normal design expectations.

8. Secondary Development Guide

8.1 Communication Protocol Summary

Protocol	Interface Type	Data Format	Reading Strategy
USB CDC	Virtual serial port, 115200-8-N-1	Pure binary stream, 64 bytes/packet	Read in blocks ≥ 64 KB, non-blocking mode
USB WinUSB	USB Bulk transfer	Pure binary stream	Batch read via libusb with large buffer

8.2 Recommended Cross-Platform Development Libraries

Language / Platform	CDC Recommended Library	WinUSB Recommended Library
Python	pyserial	pyusb + libusb
C/C++	libserialport	libusb
C# / .NET	System.IO.Ports.SerialPort	LibUsbDotNet
Node.js	serialport	usb (node-usb)
Go	go.bug.st/serial	gousb
Rust	serialport-rs	rusb
Java	jSerialComm	usb4java

8.3 Development Key Points

- 1. CDC protocol:** Open the serial port in binary mode, read at least 64 KB each time, and use non-blocking or short-timeout mode.
- 2. WinUSB protocol:** Perform Bulk IN transfers via libusb. On Windows, ensure `libusb-1.0.dll` is in the program's search path.
- 3. Data storage:** It is recommended to save read data in fixed-size blocks for subsequent randomness testing.

9. Frequently Asked Questions (FAQ)

Symptom	Solution
Windows reports missing libusb-1.0.dll	Place the DLL in the same directory as rng_WinUSB.py, or copy it to C:\Windows\System32
WinUSB device not recognised	Use the Zadig tool to install the WinUSB driver
macOS cannot find CDC serial port	Replace the data cable (not a charging-only cable); try connecting the device USB port directly
Linux reports insufficient permissions	Run <code>sudo usermod -aG dialout \$USER</code> (CDC) or configure udev rules (WinUSB)
Rate is far below the rated value	Confirm single-channel firmware is in use; close other programs occupying the port; increase the serial port receive buffer for CDC
LED always on / not lit	Re-plug to check power supply; if the abnormality persists, contact after-sales support
Data not saved to samples	Check write permission of the program directory; confirm sufficient disk space

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